

Joint Distributions and Independence

1. Joint Distribution

A joint distribution describes the probability of two or more random variables occurring together.

Example:

A = Student passes the exam

B = Student is female

$P(A \cap B)$ = Probability that a student is both female and passes.

Joint probability is used when two events happen together.

2. Independence and Dependence

Independent Events:

Two events are independent if one event does not affect the occurrence of the other.

Formula:

$$P(A \cap B) = P(A) \times P(B)$$

Example:

Tossing a coin and rolling a die.

Dependent Events:

Two events are dependent if one event affects the probability of the other.

Example:

$P(\text{Pass} \mid \text{Female})$

3. Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred.

Formula:

$$P(A|B) = P(A \cap B) / P(B)$$

Example:

$P(\text{Pass} \mid \text{Female})$

Meaning:

Probability that a student passes given that the student is female.

4. Probability Mass Function (PMF)

PMF is used for Discrete Random Variables.

Properties:

- $0 \leq P(X=x) \leq 1$
- Sum of all probabilities equals 1

Example:

Rolling a die where each outcome has probability $1/6$.

5. Probability Density Function (PDF)

PDF is used for Continuous Random Variables.

Examples:

- Age
- Height
- Salary
- Temperature

Properties:

- $f(x) \geq 0$
- Total area under the curve equals 1

Examples:

Age < 30

Salary $\leq 50,000$

6. Marginal Distribution

Marginal distribution tells us about one variable while ignoring the others.

Example:

Finding the distribution of gender without considering pass/fail status.

It provides information about a single variable only.

7. Conditional Distribution

Conditional distribution describes the distribution of one variable when another variable is known.

Formula:

$$P(X|Y) = P(X,Y) / P(Y)$$

Example:

$P(\text{Pass} | \text{Female})$

Used when information about another variable is already available.

8. Continuous Transformation of Variables

Transformation converts one random variable into another.

Examples:

- Celsius to Fahrenheit
- Monthly Salary to Annual Salary
- Centimeters to Meters

Transformations help simplify analysis and calculations.

9. Correlation vs Causation

Correlation means two variables move together.

Causation means one variable directly causes changes in another.

Important:

Correlation does NOT imply causation.

Example:

Ice cream sales and drowning incidents increase during summer. Both are related to hot weather, but one does not cause the other.

$$P(A \cap B) = P(A) \times P(B) \text{ (Independent Events)}$$

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\text{Correlation} = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$$

Quick Revision Sheet

- Joint Distribution → Probability of two events occurring together.
- PMF → Used for discrete variables.
- PDF → Used for continuous variables.
- Marginal Distribution → Studies one variable only.
- Conditional Distribution → Probability given another event.
- Covariance → Measures direction of relationship.
- Correlation → Measures strength and direction of relationship.
- Correlation range = $[-1, 1]$.
- Correlation $\neq$ Causation.
- Independent Events: $P(A \cap B) = P(A)P(B)$.